CS 391
Web Application Development

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Week - 2

Outline

CSS Basics

1. Advanced CSS Selectors
2. CSS Layout
3. What's New in CSS3?

Advanced CSS Selectors

1. Advanced CSS Selectors
2. CSS Layout
3. What's New in CSS3?

Child Selector

- Selects an element which is a **direct** child of another element.

CSS

```
ul>li {
    color: red;
}
```

HTML

```
<ul>
    <li>Apple</li>
        <ol>
            <li>Red</li>
            <li>Yellow</li>
            <li>Green</li>
        </ol>
    <li>Orange</li>
    <li>Banana</li>
</ul>
```

Result

- Apple
    1. Red
    2. Yellow
    3. Green
- Orange
- Banana

Descendant Selector

- Used to select elements which are descendants of another element in the document tree.

CSS

```
ul li {
    color: red;
}
```

HTML

```
<ul>
    <li>Apple</li>
        <ol>
            <li>Red</li>
            <li>Yellow</li>
            <li>Green</li>
        </ol>
    <li>Orange</li>
    <li>Banana</li>
</ul>
```

Result

- Apple
    1. Red
    2. Yellow
    3. Green
- Orange
- Banana

Adjacent Sibling Selector

- select an element that is **directly after** another specific element.
  - h1 + p
  - Any text appearing between markup will not affect this selector

HTML

```
<h1>Fruits</h1>
<p>Today's deals</p>
<p>30.10.2023</p>
<ul>
    . . .
</ul>
<p>Editorial recommendations</p>
```

CSS

```
h1+p {
    color:blue;
}
```

RESULT

Fruits

Today's deals

30.10.2020

- Apple
  1. Red
  2. Yellow
  3. Green
- Orange
- Banana

Editorial recommendations

General Sibling Selector

- selects all elements that are siblings of a specified element
  - h1~p
  - Any text appearing between markup will not affect this selector.

HTML

```
<h1>Fruits</h1>
<p>Today's deals</p>
<p>30.10.2020</p>
<ul>
    <li>Apple</li>
    <ol>
        <li>Red</li>
       ...
    </ol>
    ...
</ul>
<p>Editorial recommendations</p>
```

CSS

```
h1~p {
    color:blue;
}
```

RESULT

Fruits

Today's deals

30.10.2020

- Apple
  1. Red
  2. Yellow
  3. Green
- Orange
- Banana

Editorial recommendations

Attribute Selector

- Presence of attributes or their values
- Attribute Selectors
  - a[href]
    - Selects all <a> element's with an href attribute
  - input[type="radio"]
    - Selects all input elements which are of a type of "radio".
  - img[alt~="Figure"]
    - Selects all images whose attribute title contains a space separated list of values.
      - Matches <img alt="Figure 1" />, <img alt="Figure 2" />
  - html[lang|="en"]
    - Selects any element whose attribute has a value which is a hyphen separated list beginning with a specified value.
      - Matches en, en-us,en-uk.

CSS Selector Reference

- https://www.w3schools.com/cssref/css_selectors.asp

CSS Layout

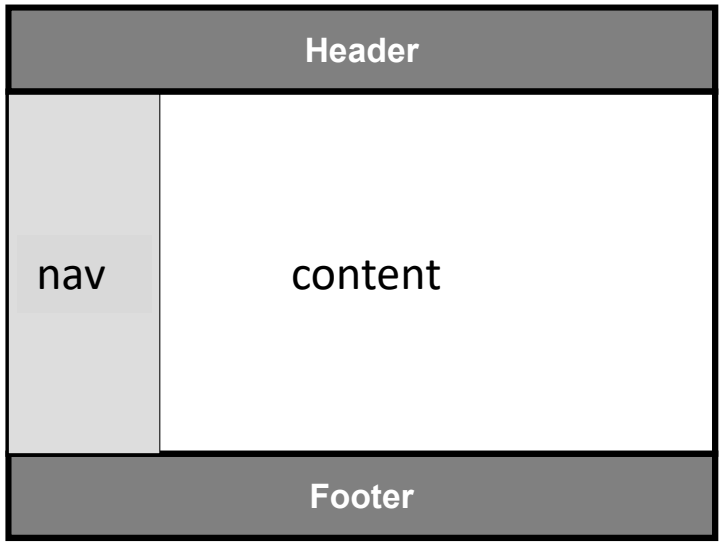
CSS Basics

1. Advanced CSS Selectors
2. CSS Layout
3. What's New in CSS3?

CSS and Tableless Layouts

- You can use CSS for tableless layouts
- Properties:
  - float
  - position
- HTML or other semantic elements (header, footer etc.)

```
<body>
   <header></header>
   <div id="middle">
      <nav></nav>
      <div id="content"></div>
   </div>
   <footer></footer>
</body>
```



The display Property

- The display property specifies if/how an element is displayed. Every HTML element has a default display value
- The default display value for most elements is block or inline.
  - none
  - inline
  - block

Example

```
li {
   display: inline;
}
```

Floating

- With CSS float, an element can be pushed to the left or right, allowing other elements to wrap around it.
- https://www.w3schools.com/css/css_float.asp

CSS

```
img
{
    float:right;
}
```

HTML

```
<img src="images/news.jpg">
<p>
        ...
</p>
```

image will float to the right in the paragraph

RESULT

A dolphin that has lived in the sea near the town of Dingle, Ireland, for nearly 40 years has gone missing. After not being seen since mid-October, there are concerns that the dolphin, called Fungie, has died or left the area. When Fungie first came to Dingle in 1983, it was a very poor area and did not have a strong tourism industry. Caroline Boland from the Dingle Peninsula Tourism Alliance told The Journal that after Fungie arrived, people started traveling there to see him.

WITHOUT FLOATING

A dolphin that has lived in the sea near the town of Dingle, Ireland, for nearly 40 years has gone missing. After not being seen since mid-October, there are concerns that the dolphin, called Fungie, has died or left the area. When Fungie first came to Dingle in 1983, it was a very poor area and did not have a strong tourism industry. Caroline Boland from the Dingle Peninsula Tourism Alliance told The Journal that after Fungie arrived, people started traveling there to see him.

Clear

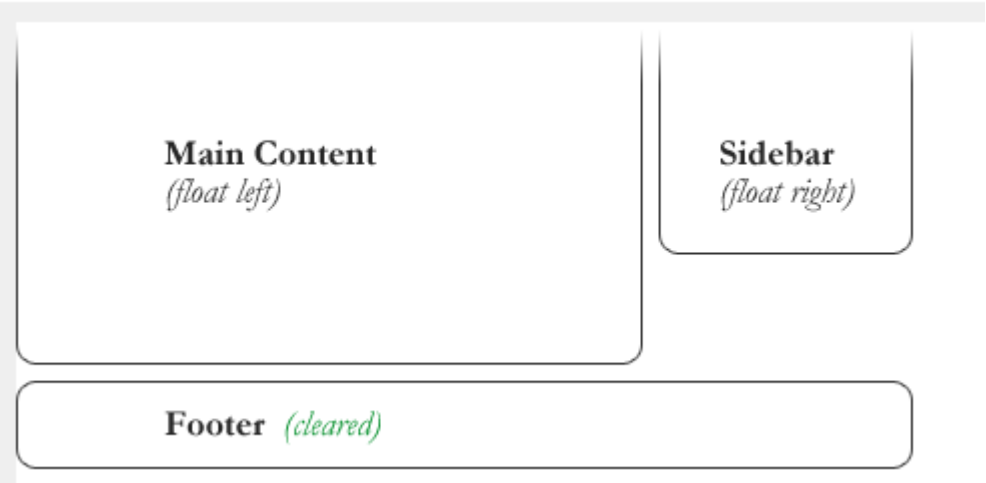
- The clear property is directly related to floats.

- The clear property specifies what should happen with the element that is next to a floating element.

- Example:
  - https://www.w3schools.com/css/css_float_clear.asp

Clear Example

without clear

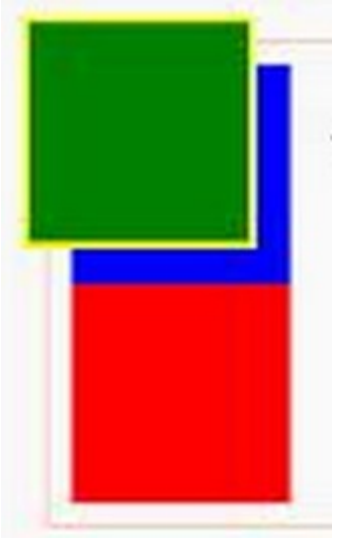


with clear

```
#footer {
   clear: both;
}
```

Positioning...

- Elements can be positioned using the top, bottom, left, and right properties.
- There are 5 different positioning methods:
  - Static
  - Fixed
  - Relative
  - Absolute
  - Sticky



Positioning (con't)

- Static
  - Element is not positioned in any special way. Position element according to the normal flow of the page. (default)
- Fixed
  - Positioning relative to the browser window
  - Element always stays in the same place even if the page is scrolled
- Relative
  - Positioning relative to its normal position
  - Behaves the same as static unless you add some extra properties: top, right, bottom, left

Positioning (con't)

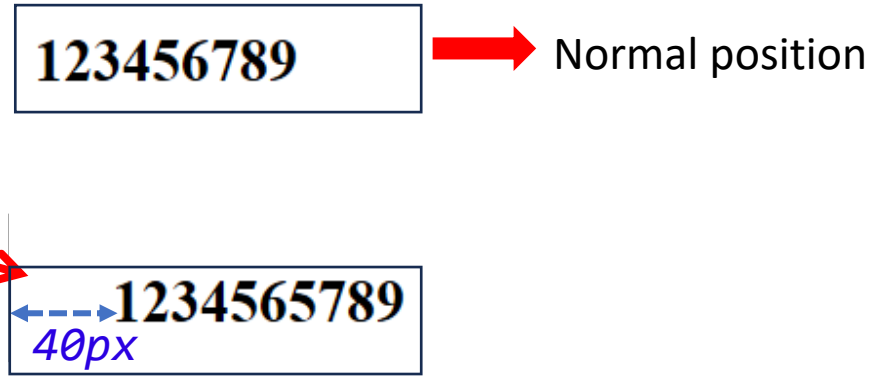
- Absolute
  - Positioning relative to the first parent element that has a position other than static
- Sticky
  - An element with position: sticky; is positioned based on the user's scroll position.
  - A sticky element toggles between relative and fixed, depending on the scroll position. It is positioned relative until a given offset position is met in the viewport - then its "sticks" in place
  - This property is only supported by all modern browsers.
  - More:
    - https://www.geeksforgeeks.org/what-is-the-difference-between-positionsticky-and-positionfixed-in-css/
    - https://www.w3schools.com/css/css_positioning.asp

Positioning Example (relative)

CSS

```
h2.posleft
{
    position:relative;
    left:40px;
}
```

WEB



HTML

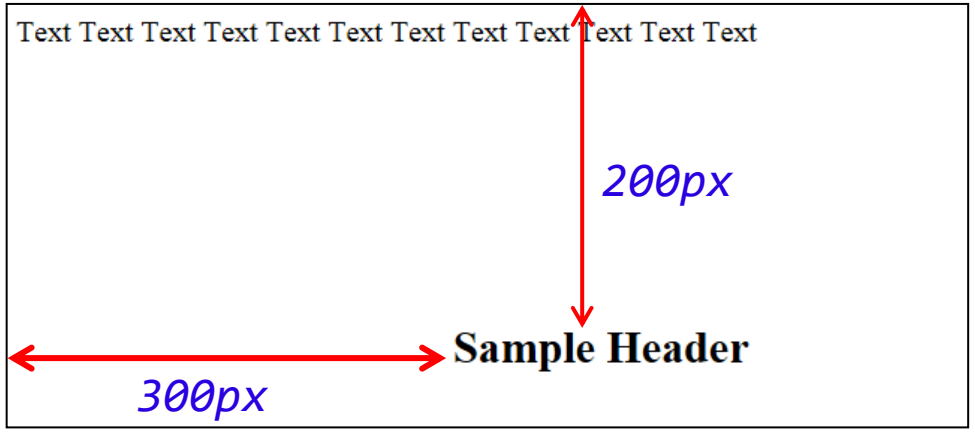
```
<h2>123456789</h2>
<h2 class="posleft">123456789</h2>
```

Positioning Example (absolute)

CSS

```
h2
{
    position:absolute;
    left:300px;
    top:200px;
}
```

window



HTML

```
<h2>Sample Header</h2>
<p>Text Text Text Text Text Text Text Text Text</p>
```

More Information for Layout

- https://www.w3schools.com/css/css_website_layout.asp
- https://www.jotform.com/blog/css-layouts-40-tutorials-tips-demos-and-best-practices/
- https://www.w3.org/2002/03/csslayout-howto
- https://websistent.com/css-tableless-design-tutorial/

What’s New in CSS3?

CSS Basics

1. Advanced CSS Selectors
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3. What's New in CSS3?

What's New in CSS3?

- CSS3 has been split into "modules". It contains the "old CSS specification". In addition, new modules are added.
- Some of the most important CSS3 modules are:
  - Selectors
  - Box Model
  - Backgrounds and Borders
  - Image Values and Replaced Content
  - Text Effects
  - 2D/3D Transformations
  - Animations
  - Multiple Column Layout
  - User Interface

CSS Animations

- An animation lets an element gradually change from one style to another
  - can replace
    - Flash animations
    - Animated images
    - JavaScripts in existing web pages.

CSS Transition

- A transition simply goes from an initial state to the final state.
  - not be a good choice a complex animation.

Transform

- A transformation is an effect that lets an element change shape, size and position.
  - Move, scale, turn, spin, and stretch elements.
  - 2D or 3D transformation

Border

- With CSS3 it is possible to
  - create rounded borders
  - add shadow to boxes
  - use an image as a border

Multi-column

- **column-count**
  Specifies the number of columns an element should be divided into
- **column-fill**
  Specifies how to fill columns
- **column-gap**
  Specifies the gap between the columns
- **…**

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Donec imperdiet tincidunt odio a blandit. Sed ornare nec turpis sit amet pretium. Praesent vulputate ligula ac purus elementum, sed luctus augue commodo. Aenean volutpat, felis vitae aliquet dictum, orci ante auctor magna, eu sollicitudin metus nisi et elit.

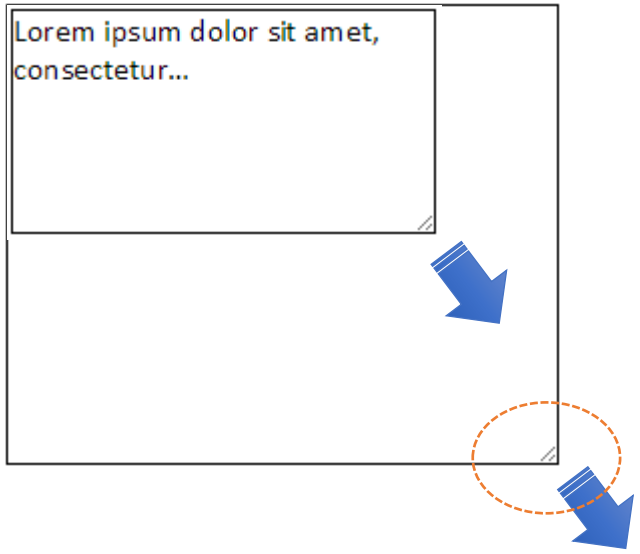
Nam bibendum tincidunt venenatis. Duis ut lacus eget arcu volutpat interdum in ac turpis. Etiam bibendum, purus eget rutrum consectetur, risus elit laoreet dui, et semper mauris turpis id leo. Morbi nulla nisi, iaculis et dictum sed, bibendum quis ante. Quisque vel varius metus.

Vestibulum tincidunt iaculis ante. Nunc vel neque leo. Pellentesque interdum, diam vitae bibendum ultrices, eros justo dignissim enim, mollis tristique odio diam vitae purus.

Resizing

```
.userinterface{
   width: 100px;
   height: 100px;
   border: 1px solid;
   resize: both;
   overflow: auto;
}
```

```
<div class="userinterface">
   Lorem ipsum dolor sit amet,
   consectetur…
</div>
```



resizing

Media Queries-Responsive Web Design

- Different layout for
  - different media types
    - such as screen or printer
  - different devices
    - Phone, tablet , etc
  - resizing browser window

For more detail wait next week

Examples

- @media (min-width: 700px) { ... }
- @media (min-width: 700px) and (orientation: landscape) { ... }
- <link rel="stylesheet" media="(max-width: 800px)" href="example.css" />
- @media screen and (min-device-width : 320px) and (max-device-width: 480px) {...}

Any Question?

Thank you for your attention

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